

Solvency Field Theory V13.9

How Two Things Become Independent: Relational Capacity, Distinguishability, and the Bell Singlet

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This replacement edition asks a narrower question than “derive quantum mechanics.” If two locally real persistent systems can share a genuinely joint phase-bearing physical relation, what preserves that relation when they separate, what physically makes them become independent, and can the same structure survive the Bell-singlet stress test? The result is a relational response model in which clean local separation preserves nonfactorization, distinguishing interactions reduce direct pair accessibility, and in the binary branch-preserving case the squared root capacity is exactly redistributed between the pair and the larger system–environment relation. The same binary response architecture gives the singlet capacity law $\mu_{st} = \frac{1}{4}(1 - st \mathbf{a} \cdot \mathbf{b})$, with no controllable superluminal signal and no preferred spacelike measurement order.

Replacement status. This document supersedes the July 2026 *SFT V13.9 Bell Narrative* and the V13.9-V2 public replacement as the controlling standalone narrative for this sector. Those documents remain historical records. The V13.9-V3 Working Relational Model v1.1 is super-frozen separately; its SHA-256 is listed in the source note.

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1 The narrow question

The original V13.9 paper asked whether a locally real picture could reproduce the Bell singlet without inserting the answer by hand. V13.9-V2 then spent much of its effort proving what *does not* work: a carried direction, a carried frame, richer source-side memory, a local branch lock, or any other independently complete local answer model remains inside Bell’s local factorization class.

V13.9-V3 changes the question. The decisive observation is that Bell is a relational problem. A single persistent object can possess invariant magnitudes, but a component along an analyzer axis, a separation, a relative orientation, or a measured spin direction exists only when there is something else for it to be relative to. The problem therefore is not “what hidden quantity does each particle carry?” It is:

What physical structure belongs irreducibly to the pair, what preserves it after the particles separate, and what interaction makes the pair become independent?

Bell remains mandatory because any proposed answer must still reproduce the quantum-strength correlation without signaling. But Bell is now a hostile regression on the relational mechanism, not the first place the mechanism is invented.

This paper also keeps one hard ontology correction. There is no physical cosmic ledger, memory surface, or universal microscopic write operator. Older “write” language is retained only as shorthand for an irreversible realized segment of four-dimensional history. The universe has nothing to write *on*; real physical support propagates and spacetime geometry responds.

2 The picture in plain language

Before the mathematics, here is the physical picture in ordinary words.

How SFT may make a pair. The leading SFT pair-production picture is currently a **strongly plausible derivation target**, not a premise of any result proved in this paper. In that picture, matter forms when open null propagation becomes organized into self-maintaining recurrent closure. In the clean isolated formation case, one photon cannot simply turn itself into one massive particle while conserving the full incoming four-momentum; a joint interaction is required. The leading candidate is therefore a two-photon event. When two photons meet with sufficient center-of-momentum energy, the combined local physical configuration may admit two persistent recurrent closures where neither incoming photon could close alone. In the present candidate picture the two closures emerge with opposite charge/framing orientation, giving an electron and a positron as charge-conjugate descendants of one event. The detailed nonlinear capture calculation—including the support scale and the branching into particular spin channels—is still a downstream derivation target.

What matters for this paper is the structural consequence, not whether that particular formation calculation ultimately wins: one local interaction can create two persistent objects whose complete physical description initially contains facts that belong to the pair rather than to either member separately.

What the shared event can leave behind. Each particle is a complete object. It can maintain its own mass, charge, and spin magnitude—properties one object can possess alone. But some questions are comparisons. “How much spin does it have?” and “which component is measured along this apparatus axis?” are not the same kind of question. The second requires another physical object that supplies the axis. A produced pair also need not automatically be a singlet. The Bell calculation below conditions specifically on the prepared $L = 0, S = 0$ channel. In that selected channel, total spin zero is a real fact about the pair. It is not a separate local label stored inside A and another matching label stored inside B. This is the sense in which V13.9-V3 treats entanglement: not as a substance, signal, or hidden memo carried by each particle, but as still-active phase-bearing structure belonging irreducibly to the joint configuration.

Why distance does not end it. Move the two particles farther apart and the fact that their joint state was nonfactorizable has not, by distance alone, been answered or erased. Once direct mechanical overlap is gone, there need be no continuing force or cable between them. Clean local evolution can simply carry the joint relation forward. Nor is there a universal clock silently counting the relation down. In the time ontology used here there is no universal elapsed time shared by the pair. A useful shorthand is: *a relation does not age; it gets spent*. “Spent” does not mean that a conserved relational substance is being consumed. It means that later physical interactions can change how much of the original joint structure remains accessible as an AB-only relation.

How two things become independent. A distinguishing interaction correlates different pair alternatives with different physical states of something else. A scattered photon can leave differently depending on the alternative;

an atom can recoil differently; a detector can amplify one distinction into a macroscopic record. The environment then carries a real physical trace of the distinction. In the clean binary case proved below, this trade is exact: as direct AB accessibility decreases, the complementary part belongs to the enlarged AB:E relation. Nothing has to be sent from A to B, and nothing has to be destroyed. What changed is which physical relation contains the distinguishability. Repeated distinguishing interactions can drive the AB-only accessibility toward zero. That, rather than distance or universal elapsed time, is the V3 mechanism by which two things become independent.

What a measurement is doing. When an apparatus measures spin “along this axis,” it is not simply reading a pre-existing analyzer-relative component carried privately by the particle. The apparatus supplies the second physical object relative to which that component is defined and recorded. The measurement therefore creates a new particle–apparatus relation while interrogating whatever larger relation the particle already participates in. Bell’s theorem together with Bell-test results excludes the corresponding measurement-independent local-answer-table class for the observed correlations. The V3 model instead asks whether the still-active joint relation, combined with two local analyzer questions, produces the observed singlet structure. The calculation below is the answer to that narrower question.

What travels between the wings: nothing. No cable, faster-than-light whisper, or preferred universal “now” is required. Each wing’s raw response distribution is independent of the distant setting, so the relation cannot be used as a signaling channel. What fails is quieter than causality: the assumption that the complete description of the pair must split into two independently complete local descriptions. In one sentence: **some physical facts belong irreducibly to relations, not to either object separately.**

The honest line. This paper does not derive which outcome becomes actual on an individual run; microscopic event realization is explicitly handed to the next program. And the exact redistribution theorem is proved only in its stated clean domain—a normalized binary pure root subjected to branch-preserving distinguishing interaction—not yet for arbitrary mixed, branch-mixing, or multipartite dynamics. Everything that follows is the mathematics of the physical picture just stated.

3 What Bell actually excludes

A measurement-independent Bell-local model has the structure

$$P(s, t \mid \mathbf{a}, \mathbf{b}) = \int d\lambda \rho(\lambda) P_A(s \mid \mathbf{a}, \lambda) P_B(t \mid \mathbf{b}, \lambda), \quad (1)$$

with $\rho(\lambda) \geq 0$ and $\int d\lambda \rho(\lambda) = 1$. Such models obey a CHSH bound $|S| \leq 2$.

The point is not that “reality” fails. Equation (1) assumes that, after conditioning on some complete source information λ , the two wings are independently complete and their joint response factorizes. Adding more local detail does not help. Thermal detector variables, carried orientations, hidden clocks, private source histories, or auxiliary local randomness can all be absorbed into a larger λ ; if the response still factorizes, the Bell bound remains.

The frozen Bell canon exhaustively enumerated the 64 deterministic local CHSH strategies and found no nonnegative mixture reproducing the singlet table. The nearest local model misses the frozen probability table by at least $1/32 = 0.03125$ in the maximum probability-cell norm. The conclusion is narrow:

causal locality can survive while Bell-local separability fails.

(2)

SFT therefore retains local source events, local analyzer interactions, causal propagation, irreversible local records, and operational no-signaling. What it gives up is the assumption that the complete physical state of an entangled preparation can be decomposed into two independently complete local states plus a common-past variable.

4 The level correction

The level structure is imported from *SFT V13.11: The Cones and the Ruler*; only the result needed here is repeated.

Level 0 contains null causal structure: light cones, causal order, conformal geometry, and null propagation. It supplies no intrinsic massive rest frame or clock.

Level 1 appears when a persistent closed process exists. One closed system can carry its own intrinsic state S_A : rest frame, local tally, and invariant magnitudes such as mass, charge, and spin magnitude.

Level 2 begins when two persistent systems exist together. Separation, relative velocity, relative orientation, spin direction, interaction, and measurement become meaningful because they are comparisons.

That hierarchy gives the central rule of V13.9-V3:

$$\boxed{\text{Do not assign to one object what physically exists only in a relation.}} \quad (3)$$

This one correction explains why the V2 search kept failing when it tried to make a single particle carry the Bell answer. A first-level object has no analyzer-relative component before the analyzer exists. The measured component is not a hidden local vector waiting to be revealed; it is a property of the particle–apparatus relation.

The time correction from *SFT V13.10: Time* is equally important. There is no universal present and no master clock shared by the two wings. Any physical account of entanglement decay that begins with “after a universal time t ” is already using a structure the theory does not grant.

5 The joint relation

Let $\mathcal{S}_A$ and $\mathcal{S}_B$ denote the complete Level-1 intrinsic states of two persistent systems. After a genuine local interaction J , their lawful joint continuation need not factor:

$$\boxed{\Omega_{AB} \neq \Gamma_A \times \Gamma_B.} \quad (4)$$

Equation (4) is already more than a common cause. Conservation alone can produce a nonproduct set of legal joint outcomes. But that is not yet quantum entanglement. Bell-relevant structure must also retain phase-sensitive information that cannot be reconstructed from two classical marginals.

The minimum V3 bookkeeping object is therefore

$$\boxed{\mathfrak{C}_{AB} = (\Omega_{AB}, \Phi_{AB}),} \quad (5)$$

where Φ_{AB} denotes physically relevant relative action/transport phase. A convenient binary chart is a complex 2×2 matrix Z , but Z is not claimed as a microscopic substance or a universal wavefunction field. It is a response coordinate for the unresolved joint configuration.

For massive alternatives, the phase source is the action difference

$$\boxed{\Delta\phi = \frac{S[\Gamma_1] - S[\Gamma_2]}{\hbar} \pmod{2\pi}.} \quad (6)$$

More generally the relative phase may contain transport or holonomy terms. A photon is not assigned a proper-time or Compton clock; native null transfer phase remains an open microscopic detail. The important point is universal: phase compares physically different four-dimensional continuations.

6 The binary response budget

The V3 binary pure response model gives one exact identity without using outcome probabilities to define its norm. Normalize the joint response chart by

$$\text{Tr}(ZZ^\dagger) = 1, \quad (7)$$

and define the local Gram matrix

$$G_A = ZZ^\dagger = \begin{pmatrix} a & b \\ b^* & d \end{pmatrix}, \quad a + d = 1. \quad (8)$$

Then define

$$P = |a - d|, \quad V = 2|b|, \quad C = 2|\det Z|. \quad (9)$$

Direct algebra gives

$$\boxed{P^2 + V^2 + C^2 = 1.} \quad (10)$$

The meanings are deliberately operational. P is local bias, V is local phase accessibility, and C is the pure binary joint factorization defect. Equation (10) says that, inside this normalized pure response chart, local and joint structure are complementary parts of one finite response budget.

The identity is exact, but its domain matters. Once AB becomes correlated with an environment, AB alone is generally mixed. V13.9-V3 does *not* extend $C = 2|\det Z|$ into a claimed general mixed-state concurrence. That would import machinery the program explicitly leaves open.

As a regression check, 10^5 random normalized complex 2×2 charts were tested numerically. The largest residual in Eq. (10) was 1.8×10^{-15} , consistent with floating-point roundoff. The numerical check validates implementation; the proof is the algebra above.

7 Why separation does not erase the relation

Suppose the pair separates after the interaction at J . Clean local propagation acts independently on each wing:

$$\boxed{Z_{ab} = L_A Z_J L_B^T} \quad (11)$$

If L_A and L_B are invertible,

$$\boxed{\text{rank } Z_{ab} = \text{rank } Z_J} \quad (12)$$

This elementary statement carries a strong physical message. Separation can remove direct mechanical overlap without turning the joint relation into a product. The particles need not remain connected by a force, cable, hidden field, or instantaneous update. The relation persists because clean local evolution does not erase the information that distinguishes a nonproduct joint configuration from a product one.

The same statement removes universal-time decay. A cleanly isolated pair does not decohere merely because “time passes.” In V13.10 there is no universal elapsed time available to perform that job. What matters is whether some later physical interaction creates a distinction between alternatives.

That distinction is the next step.

8 Distinguishability makes independence

Take a binary pure root relation in a convenient basis,

$$Z_0 = \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (13)$$

Its root pure relational capacity is

$$\boxed{C_0 = 2|\alpha\beta|} \quad (14)$$

Now let an environment distinguish the alternatives without mixing them:

$$0 \otimes e_* \rightarrow 0 \otimes e_0, \quad 1 \otimes e_* \rightarrow 1 \otimes e_1, \quad (15)$$

with normalized environment responses. Define their physical response overlap

$$\boxed{\gamma_E = \langle e_1, e_0 \rangle_G, \quad 0 \leq |\gamma_E| \leq 1} \quad (16)$$

The AB cross-response is multiplied by γ_E . The remaining *pair-accessible* relational capacity is therefore

$$\boxed{A_{AB} = C_0 |\gamma_E|} \quad (17)$$

This gives the first quantitative answer to the title question. Clean separation does not spend the relation. A physical interaction that makes the alternatives distinguishable does.

But Eq. (17) by itself still sounds as if something has simply vanished. The v1.1 freeze closes that gap.

9 The redistribution theorem

Construct the weighted Gram matrix of the two environment-conditioned responses:

$$G_E^{(w)} = \begin{pmatrix} |\alpha|^2 & \alpha^* \beta \gamma_E \\ \alpha \beta^* \gamma_E^* & |\beta|^2 \end{pmatrix}. \quad (18)$$

Its determinant is

$$\det G_E^{(w)} = |\alpha\beta|^2(1 - |\gamma_E|^2). \quad (19)$$

Define the relational capacity externalized into the enlarged AB:E configuration by

$$X_{AB:E} = 2\sqrt{\det G_E^{(w)}} = C_0\sqrt{1 - |\gamma_E|^2}. \quad (20)$$

Then

$$C_0^2 = A_{AB}^2 + X_{AB:E}^2. \quad (21)$$

Equation (21) is the **Binary Relational Accessibility Redistribution Theorem**. Its domain is specific: a normalized binary pure root relation, a branch-preserving distinguishing interaction, normalized environment responses, and a globally coherent enlarged configuration. It is not a general CKW theorem and not a universal mixed-state entanglement measure.

The physical interpretation is direct. If $|\gamma_E| = 1$, the environment has made no distinction and $A_{AB} = C_0$. If $0 < |\gamma_E| < 1$, some relation remains directly accessible to AB and some belongs to the larger configuration. If $\gamma_E = 0$, the pair-accessible term vanishes but

$$X_{AB:E} = C_0. \quad (22)$$

Nothing has been promoted to a physical fluid called “capacity.” The equation says that the mathematical signature of the root relation has moved out of the pair-only description and into the larger physical relation.

For independent repeated marks,

$$A_N^2 = C_0^2 \prod_{k=1}^N |\gamma_k|^2, \quad X_N^2 = C_0^2 \left(1 - \prod_{k=1}^N |\gamma_k|^2\right), \quad (23)$$

so

$$A_N^2 + X_N^2 = C_0^2. \quad (24)$$

The n th mark externalizes

$$\Delta X_n^2 = A_{n-1}^2(1 - |\gamma_n|^2), \quad (25)$$

and exactly

$$A_n^2 + \Delta X_n^2 = A_{n-1}^2. \quad (26)$$

The analytic identity was tested on 10^5 random normalized complex roots and random complex overlaps in the unit disk; the largest closure residual was 1.1×10^{-16} . Ten-step randomized marking sequences closed the telescoping identity to 5.6×10^{-16} . Again, the numerics are regression checks, not the proof.

This is the sharpened answer to “How do two things become independent?”:

They become independent when physical interactions progressively externalize the phase-sensitive relational accessibility that previously belonged directly to the pair.

10 Finite interface and monogamy

The redistribution theorem is exact in its binary marking domain, but general multipartite conservation is not yet derived. V13.9-V3 therefore keeps a weaker monogamy theorem that follows from finite interface capacity.

Let the relational interface of A have finite dimension d_A . Independent active relational sectors incident through that same interface obey

$$\dim \text{span}_i \mathcal{R}_{A|X_i} \leq d_A. \quad (27)$$

A maximally occupied AB interface therefore cannot simultaneously host another independent maximal AC relation using the same degrees of freedom. Apparent overcapacity must be shared, redundant, or genuinely multipartite structure rather than two independently complete pair relations.

Equation (27) is not advertised as a derivation of the Coffman–Kundu–Wootters inequality. General metric monogamy remains open. The contribution is narrower: the same finite-interface picture that supports the binary response budget also forbids independent double occupation.

11 Local questions, joint support

The Bell regression now begins. The selected source is the maximal zero-total binary relation. In the response chart,

$$\boxed{Z_{\text{singlet}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.} \quad (28)$$

The equivalent reduced source projector is

$$\Pi_0 = \frac{1}{4} (I - \boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B). \quad (29)$$

This is a preparation premise for the selected $L = 0, S = 0$ channel, not a derivation of the source branching fraction.

Each ideal binary analyzer asks one local question along an axis $\mathbf{n}$. On a two-state response space, Hermiticity, rotational covariance, axis reversal, and ideal sharpness give

$$\boxed{D(\mathbf{n}) = \mathbf{n} \cdot \boldsymbol{\sigma},} \quad (30)$$

with outcome sectors

$$\boxed{P_s(\mathbf{n}) = \frac{1}{2} (I + s \mathbf{n} \cdot \boldsymbol{\sigma}), \quad s = \pm 1.} \quad (31)$$

The analyzer is local. The source relation is joint. The actual-context response capacity for outcomes (s, t) is

$$\mu_{st} = \|P_s(\mathbf{a}) Z_{\text{singlet}} P_t(\mathbf{b})^T\|_G^2. \quad (32)$$

Direct multiplication gives

$$\boxed{\mu_{st}(\mathbf{a}, \mathbf{b}) = \frac{1}{4} (1 - st \mathbf{a} \cdot \mathbf{b}).} \quad (33)$$

The angular dependence is not supplied as a separate Bell kernel. It appears when the maximal zero-total relation is interrogated by two independently local analyzer geometries. The source object contains no future detector axes; the detector projectors contain no source-specific angular law. Their composition produces the dot product.

The mathematics in Eqs. (28)–(33) is standard two-state algebra. The claimed contribution is not invention of the Pauli matrices or the singlet identity. It is the proposed physical provenance: a Level-2 relational response model in which the same finite joint structure that persists, decoheres, and redistributes also supplies the Bell support geometry.

12 The Bell regression

The local response marginals are

$$\sum_t \mu_{st} = \frac{1}{2}, \quad \sum_s \mu_{st} = \frac{1}{2}. \quad (34)$$

Neither depends on the distant setting. At the response-capacity level, no controllable signal is available.

Define the capacity correlation

$$E_{\text{cap}}(\mathbf{a}, \mathbf{b}) = \sum_{s,t} st \mu_{st}. \quad (35)$$

Then

$$\boxed{E_{\text{cap}}(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}.} \quad (36)$$

For the standard CHSH axes,

$$\boxed{|S_{\text{cap}}| = 2\sqrt{2} \approx 2.828427,} \quad (37)$$

against the Bell-local bound $|S| \leq 2$. The violation margin is

$$2\sqrt{2} - 2 \approx 0.828427. \quad (38)$$

The old Bell canon additionally established the exact local-polytope infeasibility, delayed-choice consistency, spacelike-order independence, swapping architecture, GHZ/N-party necessity, and null-complete detection-loop-hole controls. Those results are inherited as hostile controls rather than rederived line by line here.

One conditional bridge is worth keeping because this document replaces the older public narratives. If mutually exclusive ideal completion opportunities add positively and share a common local detector rate profile $g_D(\xi)$, then

$$\lambda_i(\xi) = g_D(\xi)\mu_i. \quad (39)$$

Conditioned on completion,

$$P(i \mid \text{completion}) = \frac{\mu_i}{\sum_j \mu_j}. \quad (40)$$

For the exhaustive Bell sectors, $\sum_j \mu_j = 1$, so the ideal completed-event law is numerically the same as Eq. (33). This bridge does not derive the microscopic event generator. Why one actual event occurs is explicitly deferred to the downstream transition-dynamics program and is not required for the V3 core relational result.

13 Causal locality without Bell locality

The local analyzer maps act on separate factors and commute:

$$P_s^A P_t^B = P_t^B P_s^A. \quad (41)$$

Different Lorentz frames may reverse the coordinate-time order of spacelike detections. Equation (41) makes that ordering bookkeeping, not ontology. There is no physically preferred first measurement and no required instantaneous remote mechanical update.

The same point applies to environmental spending. Independent spacelike distinguishing interactions can be represented by commuting local maps on the enlarged relation. What changes is which part of the joint structure remains accessible to a chosen subsystem, not what a distant particle is mechanically forced to do “right now.”

The clean statement is unchanged from the original narrative:

SFT preserves causal locality but is not Bell-local.

(42)

Bell-local factorization fails because the complete prepared relation is not separable into independently complete local response tables. Causal locality survives because source formation, propagation, analyzer contact, environmental marking, and records are all local physical events, and the frozen response marginals do not depend on the remote setting.

14 What V13.9-V3 changed

The replacement from V13.9 and V2 is conceptual as much as mathematical.

First, history and current relation are separated. A shared past can explain why a relation exists, but shared history alone is only a common cause and cannot produce Bell correlations. The active relational structure belongs to the present joint configuration, not to a hidden answer table stored at formation.

Second, birth is not the only origin. Entanglement swapping shows that previously independent ancestries can become relationally joined through a later local interaction. The universal statement is therefore about joint physical configuration, not common birth.

Third, pairwise arity is not universal. GHZ-type preparations require genuine N-party relational structure; pairwise reductions need not reconstruct the whole.

Fourth, independence rather than entanglement becomes the dynamical question. Clean local separation preserves the relation. Distinguishing interaction changes accessibility. In the binary branch-preserving case Eq. (21) upgrades that statement to an exact redistribution law.

Fifth, outcome realization is removed from the critical path. The relational/Bell result does not require a hidden deterministic selector, a collapse field, a universal clock, or a solved detector avalanche. Those belong to a later program.

This last scope boundary matters. V13.9-V3 does not become a 40-page attempt to derive all transition amplitudes, cross sections, decay widths, and detector microphysics. The next program begins exactly where that physics becomes necessary.

15 The next program is different

The V3 gates that explored particle formation, detector nucleation, and a possible universal event law were useful because they localized the next problem. They are preserved as child-program prework, not promoted into the V13.9 core.

The downstream **SFT Native Transition Dynamics Program** owns the general transition map $\mathcal{T}_J$, P6/complete composition, creation and annihilation, source branching, scattering cross sections, decay widths, absolute event rates, detector completion, and general mixed/multipartite dynamics.

The V13.9 result does not depend on those being solved. Conversely, if that later program must change the frozen V13.12 isolated electron merely to make an interaction work, it has not extended the parent model; it has falsified or superseded it. *SFT V13.12: The Maintained Electron* supplies that downstream stop rule and the maintained Level-1 electron target. The only fact imported here is that the electron is represented as one recurrent locally-null maintained structure with Pauli/Gordon spin readouts; its interaction physics remains downstream.

16 Selected hostile controls

The replacement model keeps the old discipline: a physical picture must forbid something. Table 1 lists the most useful controls without reproducing the full technical canon.

Control	Result	Meaning
Bell-local deterministic strategies	64 enumerated; no nonnegative mixture	Richer local hidden detail does not reproduce the frozen singlet table.
Nearest Bell-local cell fit	minimum max-cell residual 1/32	The failure is quantitative, not semantic.
CHSH	$2\sqrt{2} \approx 2.828427$	Exact quantum-strength response-capacity regression.
Binary pure triality	max random-test residual 1.8×10^{-15}	Numerical validation of the exact algebraic identity.
Redistribution theorem	max random-test residual 1.1×10^{-16}	Binary branch-preserving capacity redistribution closes numerically to roundoff.
Repeated marks	telescoping residual 5.6×10^{-16}	Event-by-event spending identity is stable under repeated random marks.
Symmetric Bell loss control	$\eta_{\text{crit}} = 2/(1 + \sqrt{2}) \approx 0.828$	Coincidence-only CHSH is not a loophole-free experiment theorem.

Table 1: Selected hostile and numerical regression controls. Analytic identities do not depend on the random tests.

17 Claim boundary

Claim	Status	Boundary
Level-2 relational ontology	Model freeze	Relation is joint physical structure, not a carried local answer.
Joint nonfactorization	Pass	Genuine Level-2 structure; classical conservation alone is not sufficient for Bell entanglement.
Massive action phase	Pass	$\Delta\phi = \Delta S/\hbar$; native null transfer microdynamics remains partial.
Clean-separation persistence	Pass	Exact under invertible local response transport.
Finite-interface monogamy	Weaker pass	Independent relational occupation is finite; general CKW not derived.
$P^2 + V^2 + C^2 = 1$	Exact conditional	Normalized binary pure response model only; no mixed-state extension claimed.
Accessibility redistribution	Exact conditional	$C_0^2 = A_{AB}^2 + X_{AB:E}^2$ for branch-preserving distinguishing interactions.
General multipartite conservation	Open	Arbitrary branch mixing, mixed roots, and metric CKW remain downstream.
Ideal analyzer form	Conditional pass	Reduced sharp rotationally covariant binary interface.
Singlet capacity kernel	Exact conditional	Prepared maximal zero-total binary relation plus ideal analyzers.
No-signaling / spacelike order	Pass in frozen ideal model	Marginals and commuting local maps; no remote mechanical update required.
Completed Bell frequencies	Conditional bridge	$P_i = \mu_i$ under common-profile additive local completion; event generator remains open.
P6 / full quantum composition	Open	Not required for the scoped binary V3 result.
Transition dynamics	Child program	Cross sections, widths, source branching, absolute rates, detector completion.

The sentence

SFT V13.9-V3 treats entanglement as phase-bearing Level-2 joint structure rather than a property carried independently by either participant. Clean local separation preserves a nonfactorizable relation. Physical distinguishing interactions reduce the part of that relation accessible to the pair; in the binary branch-preserving case, the squared root capacity is exactly partitioned between remaining pair accessibility and the enlarged system–environment relation. The same response architecture reproduces the Bell-singlet angular law under local analyzers without a superluminal mechanical update or a universal present. General composition and actual event realization remain downstream.

Acknowledgments

The present replacement paper is the public narrative of the V13.9-V3 super-freeze developed by Shaun Fosmark and GPT-5.6 SOL. The earlier V13.9 Bell Completion Derivation Canon, the V2 closure program, the V3 gate sequence, the hostile audits, and the v1.1 redistribution theorem are absorbed here at their final claim boundaries rather than reproduced in discovery order.

Claude/Opus (Anthropic) developed the V3 restart inversion that moved the primary question from “why entanglement?” to “how do two things become independent?”, contributed cross-engine mechanism stress testing, maintained the restart charter, and proposed the vivid plain-language framing incorporated into Sec. 2 after final claim-boundary audit. Earlier V13.5 work with ChatGPT/5.5 (OpenAI) developed the contextual joint-admissibility and additive completion/hazard route that became an important historical precursor. Claude/Fable (Anthropic) contributed independent audit pressure and the broader hostile-gate methodology used throughout the current SFT research program. Earlier collaborators and archived SFT gate programs remain credited in their source documents; this replacement preserves their contributions without requiring the reader to reconstruct the historical sequence.

References and source note

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Controlling model: SFT V13.9-V3 Working Relational Model v1.1 — SUPER FROZEN
Freeze SHA-256: 422770de4a38776187e8e224db1de191acb036374ac8e4a3ab00176d40a3d51b
Public narrative status: replacement edition v1.1; V13.9 and V13.9-V2 narratives retained as historical records.